

From Replayable Traces to Provenance-Aware Rule Fusion: A Bounded Empirical Study of What RL Auditability Can and Cannot Identify

Working manuscript generated from repository evidence

September 2026

Abstract

“Replayable traces,” “causally meaningful symbols,” “shared symbolic rules,” and “composable policies” are distinct properties of a reinforcement-learning (RL) system, yet they are frequently conflated under a single label of auditability. We treat them as separate, individually testable claims and report a bounded study that measures each one against executed artifacts. The work begins from a naive prototype whose grammar, replay, and instrumentation claims did not survive scrutiny, and rebuilds the pipeline so that every statement is anchored to a checkable quantity. On two discrete control tasks (CartPole-v1 and Acrobot-v1), eight seeded agents per task share one frozen quantile state symbolizer. Exact rule overlap between two CartPole agents is high (Jaccard 0.5833, 77 of 132 shared productions), yet a fresh raw-state behavioral test yields only 51.36% action agreement (95% episode-cluster interval 50.48%–52.26%), a counterexample showing that symbolic overlap is not behavioral consensus. A provenance-aware merger composes source rule banks with explicit conflict arbitration, blind-spot detection, and fallback logging; its hash-linked decisions resolve back to source rules and replay exactly against the environment. CartPole fusion reaches 205.96 mean return versus 163.65 for the best single actor; Acrobot fusion reaches -443.37 versus -500 . Ablations expose strong, unexplained sensitivity to arbitration, including an advantage for randomized rule selection, and a fitted value-model baseline fails its quality checks. The evidence supports the *existence* of auditable trace and rule-composition mechanisms in these settings; generality, complete causal explanation, and universal auditability of arbitrary RL remain unproven. A separate designed-communication study reports a causal intervention on a learned symbol and is kept evidence-disjoint from the rule-fusion conclusion.

1 Introduction

Scenario. A trained neural policy is often opaque: given a state it emits an action, but the internal quantities that produced the choice are not directly legible. A growing body of work therefore proposes to record RL execution as symbolic structure—replayable trajectories, induced grammars, or state-action rule sets—so that a reviewer can inspect, reproduce, and reason about a policy after training. The implicit promise is that a symbolic layer makes RL “auditable.”

Limitations of existing approaches. That promise bundles several different questions, and the literature tends to answer one and claim another. First, a lossless sequence grammar that reconstructs a *seen* trace says nothing about whether its productions generalize to unseen states or

explain a decision. Second, an overlap measure over symbolic rules is a statistic about syntax, not about behavior. Third, a deterministic record can be internally consistent (a hash chain verifies) while the transitions it encodes never happened; integrity is not fidelity. These are not hypothetical worries: each one is instantiated as a failure we observed while building this system.

Problem and objective. Our objective is not to argue that an AI-native symbol system makes RL training fully auditable. It is the narrower, falsifiable question: *which specific quantities can an induced symbolic layer plus an auditable execution record actually identify, and where does the tempting inference from each quantity to a stronger one break?* We answer it by measuring each quantity separately and reporting the counterevidence where an inference fails.

Challenges. Three challenges make this harder than it appears. (i) The naive intuition that “compression equals understanding” fails at first contact: an early, readable serialization of the induced grammar made the artifact $1.80\times$ *larger* on disk than the raw trace it encoded, so symbolic reduction did not deliver a storage benefit and could not be sold as compression. (ii) Instrumentation is not free: an apparent difference between two training arms that should have been identical turned out to be a hashing-definition error, not a real effect, and had to be repaired before the equivalence claim was trustworthy. (iii) A large, precise-looking symbolic core can still fail to predict behavior, so an overlap statistic must be paired with an independent behavioral measurement.

Method. We address these by separating the pipeline into a *generator* (a stochastic REINFORCE agent that produces trajectories), a *deterministic kernel* (state symbolization, rule admission, conflict arbitration, and hash-chained logging), and a family of *instruments* (independent environment replay, a fresh-state behavioral-agreement test, and a provenance resolver) that each target one estimand. Claims are then routed through an explicit budget: positive, negative, diagnostic, and non-claim material are kept distinct.

Contributions.

1. A provenance-aware rule-fusion mechanism whose decisions record source-rule attribution, conflict arbitration, blind-spot detection, and fallback, and whose decision trace resolves back to source rules and replays against the environment (Section 4.3).
2. A shared-symbolizer, multi-seed measurement of exact rule overlap paired with an independent fresh-state behavioral-agreement test, yielding the counterexample that overlap does not identify consensus (Section 4.4).
3. A failure-driven account of how a naive symbolic audit pipeline must be corrected to support bounded claims, including an explicit pass/fail auditability taxonomy (Sections 2 and 5).
4. Arbitration and value-model diagnostics that bound the policy-composition conclusions and identify unresolved mechanisms (Section 7.3).
5. A separate causal-intervention result for learned communication symbols in a designed task, reported alongside—never merged into—the rule-fusion evidence (Section 4.6).

2 From a Naive Prototype to a Validated Instrument

This section is deliberately shallow-to-deep: it recounts the failures that shaped the final measurement design, because those failures *are* the reason the claims are bounded the way they are.

Grammar conflation. The initial prototype treated two different objects as one. A SEQUITUR-style sequence compressor, applied to a tokenized trajectory, produces a context-free grammar that can *reconstruct the string it was shown*; the source literature is explicit that this is a coding of one observed sequence, not a policy. A separate state-action rule set *state-symbol* $\rightarrow$ *action-symbol* is a candidate policy *generalization*. Merging them—and inferring rule provenance by substring tests rather than by recorded occurrences—made the prototype unable to certify either replay or generalization. The fix was architectural: the codec and the rule bank are induced, evaluated, and reported separately throughout this paper.

Compression that was not compression. A natural first reading is that a smaller symbolic grammar means a smaller artifact. It did. The readable serialization expanded a 61,295,430-byte trace into a 110,549,187-byte grammar file (1.80 $\times$), because pretty-printed string references cost more than the tokens they named. After re-encoding with integer symbol identifiers and general-purpose compression, the grammar archive was 78.2% smaller than the raw trace but still 8.4% *larger* than simply compressing the trace directly. The correct conclusion is narrow: the grammar supplies exact reconstruction and explicit production structure, not a best-in-class storage win. This becomes a permanent non-claim (Section 10).

Integrity without fidelity. A hash chain over logged records detects tampering but does not prove the recorded transitions occurred. We therefore added an *independent* environment replay check: re-execute each logged reset seed and action in the simulator and compare state, next state, reward, and the terminated/truncated flags exactly. Replay is the identification-bearing test; the hash chain is a diagnostic.

A hash bug that looked like a training difference. Two arms designed to train identically first appeared to differ. Inspection showed the comparator had folded an arm label and a grammar-sidecar counter into the hash; the weights and transitions were in fact equal. After hashing only the network state dictionary and defining a chain-normalized core-update hash that excludes non-optimization sidecar fields, the arms matched byte-for-byte. We record this correction rather than hiding it, because it is why the equivalence claim is about *optimizer-relevant* content only.

3 Formal Setup

Objects. An agent interacts with an episodic environment producing a state s_t , an action a_t , a reward r_t , a next state s_{t+1} , and terminal and truncated flags term_t and trunc_t . The observation s_t is a real vector; the action a_t is drawn from a finite set $\mathcal{A} = \{0, \dots, |\mathcal{A}| - 1\}$. A policy is a stochastic map $\pi_\theta(a \mid s)$ trained by REINFORCE with an entropy-free sampled objective and a normalized discounted return with $\gamma = 0.99$.

Definition 1 (State symbolization). A state symbolizer σ maps a real state to a tuple of bin indices

$$\sigma(s) = \left(\text{bucketize}_d(s^{(d)}; \theta_d) \right)_{d=1}^D, \quad (1)$$

where each θ_d holds the interior empirical-quantile edges of dimension d over B bins. The edges are fitted once from pooled random-policy calibration rollouts and then frozen, so that all training seeds and arms within a task share the same symbol alphabet.

Definition 2 (Admitted state–action rule). Let $N(c, a)$ be the count of observed transitions whose symbol is $c = \sigma(s)$ and action is a , and $N(c) = \sum_a N(c, a)$. The induced action for condition c is $a^*(c) = \arg \max_a N(c, a)$ (ties to the smallest index), with support $\kappa_s(c) = N(c, a^*)$ and confidence

$$\kappa(c) = \frac{N(c, a^*(c))}{N(c)}, \quad \text{rule } c \rightarrow a^*(c) \text{ is admitted iff } \kappa_s(c) \geq \kappa_s^{\min} \wedge \kappa(c) \geq \tau. \quad (2)$$

The admitted rules form a rule bank $\mathcal{R}$. Each rule stores the multiset of source step identifiers that produced it, and the digest of that multiset.

Equation (2) uses a single normalization for confidence; support and confidence are reported separately from coverage so that a precise-but-rare rule is never described as broadly applicable.

4 Mechanisms

4.1 A grammar-constrained policy

Given an active bank $\mathcal{R}$ from Definition 2, a constrained policy masks the action set to the rule’s action when the current symbol is covered and leaves it unconstrained otherwise:

$$\mathcal{M}(c) = \begin{cases} \{a^*(c)\}, & c \in \mathcal{R}, \\ \mathcal{A}, & c \notin \mathcal{R}, \end{cases} \quad \pi_c(a \mid s) = \frac{\exp z_a(s) \mathbf{1}[a \in \mathcal{M}(\sigma(s))]}{\sum_{a' \in \mathcal{M}(\sigma(s))} \exp z_{a'}(s)}. \quad (3)$$

The mask is a *state-conditioned* restriction with an explicit unconstrained fallback, which corrects an earlier global action-pruning design that could permanently remove an action merely because it was absent from frequent rules.

4.2 Hash-chained records and replay validity

Each logged event e_t commits to its predecessor:

$$h_t = \text{SHA256}(h_{t-1} \parallel \text{canon}(e_t)), \quad h_{-1} = \text{null}, \quad (4)$$

so modifying any e_t breaks every later digest. Integrity of Equation (4) is a diagnostic. The identification-bearing predicate is *environment replay validity*: a trace is replay-valid iff, re-executing recorded reset seeds and actions in the simulator, every step’s state, next state, reward, and both terminal flags match the record exactly (compared as 32-bit floating-point quantities and Boolean flags). A hash chain can be internally consistent while replay fails; only the latter supports a transition-fidelity statement.

4.3 Provenance-aware rule fusion

At a state s with symbol $c = \sigma(s)$ and K source rule banks $\mathcal{R}_1, \dots, \mathcal{R}_K$, the merger forms the candidate set

$$\mathcal{C}(c) = \{i : c \in \mathcal{R}_i\}, \quad \text{covered} \iff |\mathcal{C}(c)| \geq 1, \quad \text{conflict} \iff |\{a_i : i \in \mathcal{C}(c)\}| > 1. \quad (5)$$

Covered non-conflicts and agreements emit the common action; conflicts are resolved by the declared priority order (confidence, support, mean reward, –seed); uncovered states take an actor fallback. Each decision records both candidate rule identifiers, their full statistics and source-step digests, and the selected source or fallback, and enters a chain of the form of Equation (4). A validator resolves every selected reference back to its source bank, re-checks the recorded statistics and emitted action, verifies the chain, and replays the resulting actions in the environment. Blind-spot selection uses training-time returns only, so no evaluation return leaks into the fallback choice.

4.4 Behavioral agreement and symbolic overlap

For two frozen actors π_1, π_2 and a fresh held-out state distribution $\mathcal{D}$ sampled equally from each actor’s own occupancy, define

$$A = \Pr_{s \sim \mathcal{D}} [\arg \max_a \pi_1(a | s) = \arg \max_a \pi_2(a | s)], \quad (6)$$

estimated on 10,000 fresh raw states and bracketed by an episode-cluster bootstrap. The symbolic-overlap statistic is exact-set Jaccard over (state-symbol, action-symbol) productions of the two banks,

$$J = \frac{|\mathcal{R}_1 \cap \mathcal{R}_2|}{|\mathcal{R}_1 \cup \mathcal{R}_2|}. \quad (7)$$

Equations (6) and (7) are measured on the same agents precisely so that one cannot be substituted for the other.

4.5 Trajectory coding length

The lossless codec induces pair-substitution productions and reports a declared two-part description length

$$L_{\text{mdl}} = \underbrace{L(\mathcal{R}_{\text{cfg}})}_{\text{grammar}} + \underbrace{L(\tau | \mathcal{R}_{\text{cfg}})}_{\text{start sequence}}, \quad (8)$$

with Elias-gamma-coded counts and literal terminal dictionary, and fixed-width symbol references; provenance bytes are reported separately. Equation (8) is a description-length score under a stated convention, not an optimization objective.

4.6 Causal symbol intervention (separate sub-study)

In an independent two-agent private-target coordination game, a receiver maps a state and a discrete sender message to an action. The matched-support intervention contrasts the receiver’s action distribution under the observed message m against a counterfactual message m' that keeps the information-bearing symbol’s context fixed:

$$\Delta = \mathbb{E}_{s \sim \mathcal{S}} \left[\sum_a \pi_R(a | s, m) - \pi_R(a | s, m') \right], \quad (9)$$

evaluated only on the common support of the two conditioned state sets. Equation (9) is computed for the AIM sub-study alone and is not merged into the rule-fusion estimands.

5 The Instrument and What It Identifies

We separate a set of quantities that a single “auditability score” would collapse (Table 1). Trace integrity (Equation (4)) is *tamper-evidence*: a counterfactual in which a modified record still passes is unreachable by construction and so is not identification-bearing. Environment replay is identification-bearing but only for the recorded executions under the pinned software and hardware configuration. Lossless coding (Equation (8)) is a reconstruction statement. Rule coverage and agreement measure how often a rule applies and how often it matches a source actor. Behavioral agreement (Equation (6)) is an independent action-level estimand. Composition quality (Equation (5)) is a measured return under declared rules. Value-model reliability is a separate gate the fitted critics in this study do not pass.

Quantity	Estimand / predicate	What it cannot establish
Trace integrity	Chain (4) verifies	transitions occurred; logger honesty
Environment replay	exact transition match	cross-hardware equivalence
Lossless coding	reconstruction (8)	policy explanation; storage win
Rule coverage/agreement	support/confidence (2)	generalization at low coverage
Behavioral agreement	A in (6)	optimality; causal rationale
Symbolic overlap	J in (7)	behavioral consensus
Composition quality	return of (5)	optimal arbitration

Table 1: Auditability disaggregated into separately measured quantities. Each row’s right column is a boundary enforced by a Phase-0 negative or diagnostic result.

Transport versus generation. For the AIM sub-study, the intervention changes the message content presented to the receiver, not its transport; it is therefore classified as a generation-relevant change. For the RL kernel, the merger neither samples nor learns; it is a deterministic decision procedure. A change of record serialization is never treated as a change of policy.

6 Experimental Design and Registered Predictions

Eight seeds {11, 29, 43, 71, 101, 149, 211, 307} train REINFORCE agents on CartPole-v1 and Acrobot-v1 for 300 episodes per seed and arm, under one frozen shared quantile symbolizer per task ($B = 4$ bins, calibration rollouts pooled across seeds). Rule admission uses support $\kappa_s^{\min} = 8$ and confidence $\tau = 0.70$. A baseline arm and an audit-only observer arm are checked for exact optimizer-level equivalence. Composition uses 100 common reset seeds per policy; Acrobot arbiter ablations reuse those seeds, and randomized rule selection repeats over five independent selector streams. A separate five-stream uniform-random-action control isolates unconditioned action noise from state-conditioned rule availability. Stochastic return summaries use the across-episode standard deviation with a percentile bootstrap over episodes; state-level agreement treats sampled states as the measurement unit but clusters the bootstrap by episode.

Four hypotheses were recorded before the corresponding measurements:

- H1** A grammar induced from training symbols reconstructs those trajectories exactly, but held-out episodes are not “replayed” merely because their tokens are in the alphabet.
- H2** A learned state–action rule bank predicts an independent holdout above an action-frequency baseline, with coverage, agreement, and confidence reported separately.
- H3** Grammar constraints applied only in well-supported states do not harm return when paired over seeds, with an explicit low-coverage fallback.
- H4** Reconstructing a sampled trace alone is insufficient for whole-training auditability; configuration, transitions, update boundaries, and checkpoints must be recorded and re-derived.

Each prediction is scored in Section 7 as supported, partially supported, or contradicted. Because the measurements were exploratory in the final shared-symbolizer batch, the H-statuses are reconciliations against pre-specified statements rather than a formal preregistration.

Task	Policy	Mean return	Paired Δ vs best	95% CI
CartPole	best single actor	163.65	—	—
CartPole	actor-logit ensemble	500.00	—	—
CartPole	grammar fusion	205.96	+42.31	[23.98, 61.93]
Acrobot	best single actor	−500.00	—	—
Acrobot	actor-logit ensemble	−500.00	—	—
Acrobot	grammar fusion	−443.37	+56.63	[39.32, 75.15]

Table 2: Composition returns over 100 matched reset seeds. CartPole is ceiling-saturated, so the ensemble comparison is not a controlled superiority test.

7 Results

7.1 Overlap versus behavior (H2)

Prediction. H2 anticipated that a large shared rule core would coincide with faithful cross-seed behavior. *Result.* It does not. Across eight agents, mean pairwise exact overlap is $J = 0.5572$ on CartPole and 0.0956 on Acrobot. For seeds 11 and 29, CartPole has 77 shared productions out of a 132 union ($J = 0.5833$) with no action conflicts on shared conditions; Acrobot has 21 of 423 ($J = 0.0496$) with 77 of 98 shared conditions in action conflict. On 10,000 fresh CartPole raw states, the seed-11 and seed-29 actors agree on 51.38% of argmax actions; the episode-cluster balanced estimate is 51.36% (95% interval 50.48%–52.26%), near chance for a two-action task. This is a central negative result and **contradicts** H2: a substantial exact intersection does not imply behavioral consensus, and J alone is disqualified as evidence of a shared policy core.

7.2 Trace and provenance auditability (H1, H4)

For the recorded artifacts, chain, update-ledger, provenance-resolution, and environment-replay checks pass under the audited setup. The eight-seed records span 1,725,655 CartPole and 2,544,513 Acrobot trace rows across three arms; replay covers 1,122,403 and 1,742,570 transitions respectively. Every logged prefix round-trips exactly (Equation (8)), and a tamper probe that edits a single reward is rejected at the first line. These support **H1** and a bounded **H4**: exact replay is evidence about recorded executions and specified software/hardware conditions; it does not establish that the logger captured every training cause, prove logger honesty, or guarantee cross-hardware equivalence.

7.3 Composition returns and arbitration sensitivity (H3)

Prediction. H3 anticipated that well-supported constraints with a fallback would not harm return. *Result.* The online majority constraint is unstable: in the two-seed CartPole pilot, the constrained-arm return changed by −378.7 for seed 11 but +31.5 for seed 29 (mean −173.6, disagreeing signs); lowering the threshold to 0.70 gave −330.4 and +17.0. This **contradicts** H3 as a performance claim and motivates reporting composition returns as descriptive only.

Table 2 gives the shared-symbolizer composition returns. CartPole fusion reaches 205.96 versus 163.65 for the best single actor (paired difference +42.31, 95% bootstrap [23.98, 61.93]), but some policies hit a 500-step ceiling that masks differences. Acrobot fusion reaches −443.37 versus −500 (paired +56.63, 95% [39.32, 75.15]) at 95.70% rule coverage, 89.24% conflict, and 4.30% blind spots; 39 of 100 episodes terminate early, though the median remains −500.

Arbiter ablations (Table 3) show strong, unexplained sensitivity. The declared order on confidence, support, and mean reward scores −443.37; confidence-only −444.49 (paired −1.12, 95%

Acrobot arbiter variant	Mean return	Paired Δ vs declared
confidence/support/reward (declared)	−443.37	—
confidence-only	−444.49	−1.12
mean-reward-only	−500.00	−56.63
support-first	−500.00	−56.63
defer conflicts to ensemble	−500.00	−56.63
random rule selection (5 streams)	−198.82	+244.55
uniform random action (5 streams)	−498.71	−55.34

Table 3: Matched-seed arbitration ablations. Random rule selection beats the declared arbiter; random action does not, so the effect tracks state-conditioned availability, not action noise.

[−3.16, +0.06]); mean-reward-only, support-first, and conflict deferral to the ensemble each score −500. Randomly choosing among available source rules scores −187.02 on one stream and −194.10 to −206.17 across five (mean −198.82), while a five-stream uniform-random-action control averages −498.71. The mean-reward key is near-inert here (1575 of 1694 Acrobot rules have mean reward exactly −1.0). The random-action control separates state-conditioned rule availability from unconditioned noise, but the mechanism of the randomized-selection advantage is *not* identified, and the tested confidence-ranked arbiter is not shown to be a strong conflict resolver.

7.4 Value-model diagnostics (kept separate)

The fitted value baseline fails its quality gates and is reported apart from the fusion result. On CartPole it achieves mean return 9.34 (versus 163.65 for the best actor) and matches the actor-logit ensemble on 35.27% of 31,563 held-out states. On Acrobot it scores −499.57 and selects action 0 on all 50,000 diagnostic states. On 200 Acrobot states the seed-11 and seed-29 critics correlate at 0.652 in scalar value while their greedy actions diverge; fusion matches the seed-11 argmax on 64.5%, the seed-29 argmax and multi-critic improvement on 0%, and the best first action under a fixed ensemble continuation on 22.5% (95% interval 16.5%–28.5%). A scalar-value correlation is not action agreement; these critics are an unvalidated comparator, not evidence against the grammar-fusion return.

7.5 Causal symbol intervention (separate sub-study)

Prediction. None registered for the rule-fusion estimator. *Result.* Across six seeds and 600,000 logged decisions in the designed coordination game, sender-sampled receiver accuracy averages 0.998923, versus 0.503292 for shuffled messages and 0.500 for a no-message control. The matched-support intervention (9) on the information-bearing symbol changes receiver action probability by 0.998366 on average (per-seed 0.996986–0.999685; mean common-support overlap 0.94308), and the active token position varies across seeds. This supports a causal role for a learned symbol *in the tested game*; it does not provide a general semantic decoding of the language, and its executed artifacts live outside the compact review package, so it is kept evidence-disjoint from Sections 7.1 and 7.3.

7.6 Pass/fail summary

Identification-bearing outcomes (replay fidelity; the overlap-versus-behavior counterexample) are reported apart from operational-health rows (arm equivalence), calibration/diagnostic rows (hash chain, lossless coding), and the failed value gate.

Criterion	Estimate	Threshold	Verdict	Layer
Replay match (CartPole)	1,122,403 transitions	exact	pass	identification
Replay match (Acrobot)	1,742,570 transitions	exact	pass	identification
Baseline/observer equivalence	byte-identical core	exact	pass	health
Lossless reconstruction	ratio 0.513–0.575	round-trip	pass	diagnostic
Storage vs. generic compression	+8.4%	smaller	fail	diagnostic
Behavioral agreement A	51.36%	$\gg$ chance	fail	identification
Symbolic overlap J (CartPole)	0.5833	descriptive	reported	descriptive
Value-model gate	action 0 everywhere	informed	fail	diagnostic

Table 4: Consolidated verdicts. The single identification-bearing failure (behavioral agreement) and the identification-bearing passes (replay) are separated from health and diagnostic rows.

8 Related Work

Direction: interpretable and programmatic policies. Programmatically interpretable RL represents policies in a domain-specific language and reasons about them symbolically. Our setting differs: rules are induced from logged traces, so provenance and predictive generalization must each be earned, not assumed. That requirement is exactly where our negative result bites—syntax overlap does not grant behavioral faithfulness.

Methodology: sequence structure and policy composition. Linear-time grammar induction identifies hierarchical structure in sequences but, by the original authors’ own caveat, codes a single observed string rather than a generalized policy. We keep that distinction operational (Definition 2 versus Equation (8)). Generalized policy improvement bounds a composed policy by the maximum per-state advantage of its sources; successor-feature composition likewise presumes reasonably fitted value functions. Our fitted critics violate that premise (Section 7.3 diagnostics), so we invoke the composition *idea* only as a diagnostic baseline, not as a bound we establish.

Measurement of emergent communication and reproducibility. Pitfalls in measuring emergent communication motivate treating a designed-game causal test and a state–action rule test as separate estimands. Deterministic-implementation analysis for deep RL separates deterministic code paths from same-condition replicability, which is why our replay claim is bounded to the recorded host and versions rather than generalized. Where these lines of work and ours could conflict—for instance on whether symbolic overlap should predict behavior—our evidence is a single-task counterexample, so we mark it as a bounded observation, not a refutation of a broad claim.

References

- [1] Verma, Murali, Singh, Kohli, and Chaudhuri. Programmatically interpretable reinforcement learning. *ICML*, PMLR 80, 2018.
- [2] Nevill-Manning and Witten. Identifying hierarchical structure in sequences: a linear-time algorithm. *Journal of Artificial Intelligence Research*, 7:67–82, 1997.
- [3] Thakoor, Rowland, Mavrin, Borsa, Dabney, Munos, and Barreto. Generalised policy improvement with geometric policy composition. *ICML*, PMLR 162, 2022.

- [4] Barreto, Dabney, Munos, Hunt, Schaul, van Hasselt, and Silver. Successor features for transfer in reinforcement learning. *NeurIPS* 30, 2017.
- [5] Lowe, Foerster, Boureau, Pineau, and Dauphin. On the pitfalls of measuring emergent communication. *AAMAS*, 2019. arXiv:1903.05168.
- [6] Nagarajan, Warnell, and Stone. Deterministic implementations for reproducibility in deep reinforcement learning. arXiv:1809.05676, 2018.

9 Discussion

Why the mechanism can be trusted as far as it is. The audit capability rests on two equations that are independently checkable: the chain commitment (4) makes tampering detectable, and the replay predicate makes transition fidelity verifiable in the environment. The merger’s decisions are auditable because each carries its source references, which resolve back to the inducing statistics. None of this depends on the grammar being an explanation; it depends on the record being reconstructible and re-executable.

Expected versus unexpected outcomes. Expected: replay fidelity, high rule coverage, and the value-model fragility. Unexpected: (i) a large exact overlap coexisting with near-chance behavior (51.36% at $J = 0.5833$), and (ii) randomized rule selection beating the confidence-ranked arbiter by well over two hundred return points. For (i) the alternative explanation is that many distinct rule sets induce the same, and different, argmax behaviors once states are quantized differently; our data cannot separate this from genuine divergence. For (ii) a plausible alternative is that the advantage comes from the stochastic mixture escaping the -500 truncation trap, since candidate availability is state-conditioned—but the uniform-action control (-498.71) shows it is not unconditioned noise alone. We leave the mechanism unidentified.

Implications. Theoretically, the results caution against treating a syntax statistic as a semantic or behavioral one. Practically, they suggest that provenance-carrying composition is inspectable without being optimal. Neither implication claims downstream deployment quality we did not measure.

Steelman. The strongest objection is that the headline fusion gain is an artifact of a saturated comparison: both best-single and ensemble sit at the Acrobot -500 cap, so almost anything that varies actions with the state will appear better, and the randomized-selection result makes the “intelligent” arbiter look incidental. We take this seriously: we do not claim the arbiter is a good conflict resolver, we report that random beats it, we disclose the ceiling on CartPole, and we keep the improvement language descriptive of these policies and reset seeds. What survives the objection is smaller but real: the composition trace is auditable and replayable, and its failure mode (overlap $\neq$ behavior) is measured directly rather than inferred.

Replication (bounded to the repository). Independent replication requires the full multi-gigabyte trace and checkpoint tree; the compact review package supplies the summaries, overlap tables, arbitration ablations, the random-action control, Experiment B, and the value diagnostics

with recorded digests. Minimum conditions are: the pinned interpreter, tensor, and environment-library versions; deterministic single-thread CPU settings; the shared calibration quantiles; and re-derivation of every reported digest. Replaying a recorded output is distinguishable from generating a fresh one; only the latter tests transfer.

10 Limitations and Future Work

Non-claims. We do not claim: general or complete auditability of arbitrary RL training; that symbolic overlap entails behavioral consensus; that induced grammars are human-readable explanations of a neural policy; a storage-compression advantage over generic compression; optimal or mechanism-explained arbitration; validated value-model composition; or a natural-language semantics for the communication symbols. **Boundaries.** Evidence covers two benchmark tasks, eight seeds per task, one shared quantizer per task, and one software/hardware configuration; CartPole saturates and many Acrobot episodes remain at -500 ; fresh-state behavior is measured for the seed-11/29 CartPole pair, not every pair or task. **Future work** follows directly from the unresolved mechanisms: decompose the random-arbitration advantage by source rule, action frequency, and state occupancy; test return-weighted rule-bank induction under a pre-specified normalization and held-out evaluation; use the native sequence symbols as the control representation; and evaluate behavior on non-saturated tasks and across hardware.

11 Conclusion

In the evaluated prototype, executed evidence supports replayable trace inspection, provenance-aware rule composition, explicit conflict and blind-spot reporting, and a causal symbol effect in a designed communication game. The same evidence shows that grammar overlap did not predict actor agreement, that arbitration behavior remains unexplained, and that a fitted value baseline failed its checks. We therefore claim bounded existence, not universality: these mechanisms make specified RL artifacts more auditable under tested conditions, while general auditability and complete causal explanation remain open.